

Software Project Management



Project Details

Project Title:	SalesPoint
Project details:	Sales and Inventory Management System for General Retail Store
Submitted To:	Dr. Waqar Mehmood
Course:	Software Project Management
Submission Date:	06/22/2026

Group Members

Name	SAP ID
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1. Project Description:

- **Project Title:** SalesPoint (Sales and Inventory Management System for a General Retail Store)
- **Project Overview:** SalesPoint is a desktop-based Management Information System designed for small and medium-sized retail stores that sell grocery items, beverages, and daily household products. The system replaces paper-based sales and inventory records with a Java Swing application connected to a MySQL database through JDBC. It supports secure login, product and employee management, real-time stock updates, sales processing, invoice generation, customer records, and sales reporting.
- **Final Technology Stack:** Java Swing for the graphical user interface, Java business logic, JDBC for database connectivity, MySQL for data storage, and Windows desktop computers for deployment.
- **Development Methodology:** A hybrid approach is used. The scope, budget, and eight-week baseline are planned in advance, while development and testing are organized into short Agile sprints using Planner, Taiga, and Jira boards.
- **Project Objectives:**
 - Automate core sales and inventory operations of a general retail store.
 - Reduce billing and stock-record errors by at least 90% compared with manual calculations.
 - Update inventory within two seconds after a completed sales transaction.
 - Generate an accurate printable invoice within five seconds.
 - Provide daily and monthly sales reports for business analysis.
 - Maintain secure and searchable product, customer, employee, and sales records.

2. Stakeholder List:

Stakeholder	Role	Interest / Responsibility
Store Owner	Primary Client	Profit, business control, reports, final approval
Admin	Primary System User	Product, employee, inventory, and report management
Sales Staff	Secondary System User	Processing sales and generating invoices
Customers	External Stakeholders	Accurate billing and faster service
Project Team	Developers / Analysts	Design, development, testing, and documentation

3. Problem Statement:

- **Problem Description:** Many small retail stores still rely on registers, handwritten invoices, and manually maintained stock records. These methods are slow, difficult to audit, and vulnerable to calculation mistakes and record loss.
- **Current Issues in the Existing System:**
 - Manual entry of sales transactions and product quantities.
 - Errors in billing totals, stock counting, and sales calculations.
 - Difficulty in tracking daily and monthly business performance.
 - No automatic inventory deduction after a sale.
 - Risk of damaged, missing, duplicated, or outdated records.
 - Slow invoice preparation and limited accountability for staff actions.
- **Limitations of Manual Processes:** Physical records require storage space and are difficult to search. Inventory is not updated immediately, which may cause overstocking, stock shortages, or the sale of unavailable products. Manual reports require repeated calculations and may produce inaccurate results.
- **Expected Improvement through Automation:** SalesPoint will process sales through a controlled workflow, calculate totals automatically, reduce stock after each completed transaction, print invoices, and generate reports from the stored data. The system will improve operational efficiency, record accuracy, traceability, and decision-making.

4. Feasibility Analysis:

- **Technical Feasibility:** The selected technologies are suitable for the team's skills and the project scope. Java Swing supports desktop user interfaces, MySQL provides structured relational storage, and JDBC provides reliable communication between the application and database. Development can be completed using freely available tools such as JDK, NetBeans or IntelliJ IDEA Community Edition, and MySQL Community Server.
- **Economic Feasibility:** The total project budget is PKR 520,000. The expected first-year benefit is PKR 2,300,000 through labour-time savings, fewer billing mistakes, reduced stock losses, and improved reporting. The expected net benefit is positive.
- **Operational Feasibility:** The system follows familiar retail workflows and uses clear forms, tables, buttons, and confirmation messages. A two-hour training session and a user guide will support store staff adoption.
- **Schedule Feasibility:** The final schedule is eight weeks, or 40 working days. The schedule is aligned with the three-person team, the 2,400 LOC estimate, and the COCOMO effort of approximately six person-months.

5. Project Scope:

In-Scope Features:

- Secure admin login and role-based access.
- Product add, update, delete, search, and stock monitoring.
- Customer and employee record management.
- Sales transaction processing and automatic stock deduction.
- Printable invoice generation.
- Daily, monthly, stock, employee, and customer purchase reports.
- Database backup, restore verification, user training, and project documentation.

Out-of-Scope Features:

- Online shopping and customer self-service portal.
- Online payment gateway and delivery tracking.
- Mobile application.
- Multi-branch synchronization and cloud hosting.
- Advanced AI forecasting and barcode hardware integration.

Assumptions and Constraints:

- The application will run on Windows computers connected to a local MySQL database.
- The store will provide sample product, employee, and sales data for testing.
- The project must be completed within eight weeks and PKR 520,000.
- The prototype supports one retail store and a limited number of concurrent users.

6. Resource Planning:

Resource Type	Description	Allocation
Human	Project Manager, Technical Analyst / Documentation Coordinator, Java Developer / Tester	Part-time academic project team for eight weeks
Hardware	Development laptops, test computer, receipt printer, external backup drive	Existing laptops plus shared testing devices
Software	JDK, Java Swing, MySQL, JDBC, IDE, MS Project, Planner, Taiga, Jira	Community tools and institutional licences
Data	Sample products, customers, employees, and sales records	Prepared for development, testing, and UAT

7. Time Planning (Final Project Schedule):

Phase	Schedule	Main Activities
Phase 1: Planning, Requirements, and Feasibility	Week 1	Project planning, requirement gathering, feasibility, scope approval
Phase 2: System and Database Design	Week 2	Java Swing wireframes, navigation, MySQL schema, design review
Phase 3: Core Development	Weeks 3-5	Authentication, product, inventory, sales, invoice, reports, customer, employee modules
Phase 4: Integration	Week 6	JDBC integration, module integration, data validation, error handling
Phase 5: Testing and UAT	Week 7	System testing, regression testing, defect fixing, client UAT
Phase 6: Deployment and Training	Week 8	Installation, backup test, user training, documentation, final acceptance

Final baseline: 06 May 2026 to 16 June 2026, consisting of 40 working days.

8. Cost Planning:

Cost Category	Details	Amount (PKR)
Human Resources	Six person-months: Ahmad - project management; Atia - analysis and documentation; Zain - Java/MySQL development and testing	300,000
Hardware & Peripherals	Testing computer support, receipt printer, external backup drive, and maintenance allowance	120,000
Tools & Software	MS Project/Office access, design utilities, documentation, and installation support	20,000
Testing & Contingency	Test data preparation, QA support, defect buffer, and unexpected technical costs	50,000
Training & Rollout	User guide, training material, installation, and two-hour staff training	30,000
Total Project Budget		520,000

- **Basic Cost-Benefit Analysis:**

Benefit Source	Expected First-Year Benefit (PKR)
Labour-time and faster transaction processing	840,000
Reduction in stock shortages, overstocking, and record loss	720,000
Reduction in billing and calculation errors	360,000
Improved reporting and business decisions	380,000
Total Expected First-Year Benefit	2,300,000

- Total Cost = PKR 520,000
- Net Benefit = PKR 2,300,000 - PKR 520,000 = PKR 1,780,000
- ROI = (Net Benefit / Total Cost) x 100 = 342.3%
- Estimated Payback Period = PKR 520,000 / (PKR 2,300,000 / 12) = approximately 2.7 months

9. Team Planning:

- **Project Manager / Team Leader - Ahmad Raza:** Project coordination, client communication, baseline approval, schedule and budget control, risk monitoring, and final acceptance.
- **Technical Analyst & Documentation Coordinator - Atia Rahat:** Requirement gathering, process analysis, UI and database design support, test-case preparation, UAT coordination, meeting records, and final documentation.
- **Java Developer & Tester - Zain Zaheer:** Java Swing interface development, Java business logic, MySQL database implementation, JDBC integration, unit testing, defect fixing, installation, and technical support.

10. Risk Planning:

Risk ID	Risk Description	Probability	Impact	Response Strategy / Owner
R1	Requirement changes after approval	Medium	High	Use signed baseline and formal change requests. Owner: Ahmad
R2	Team member unavailability	Low	Medium	Share code and documents; maintain task backup. Owner: Ahmad
R3	Schedule delay	Medium	High	Weekly tracking, milestone reviews, and three-day non-critical-path slack. Owner: Ahmad

Risk ID	Risk Description	Probability	Impact	Response Strategy / Owner
R4	Java/MySQL integration issue	Medium	High	Build early JDBC prototype and test database connection in Week 2. Owner: Zain
R5	Data loss or database corruption	Low	High	Daily local backup and verified restore procedure. Owner: Zain
R6	Incorrect invoice or stock deduction	Medium	High	Automated calculations, validation, unit tests, and regression tests. Owner: Atia & Zain
R7	User resistance or training difficulty	Low	Medium	Simple UI, user guide, demonstration, and two-hour training. Owner: Atia

11. Monitoring & Control Planning:

- **Monitoring Frequency:** Weekly project review and daily team updates during development and testing.
- **Performance Indicators:**
 - o At least 90% of planned tasks completed by their milestone date.
 - o Schedule variance kept within three working days.
 - o Budget variance maintained below 10%.
 - o At least 95% of planned test cases passed before release.
 - o Zero open critical or high-severity defects at deployment.
 - o All approved changes recorded in the change log.
- **Reporting:** Atia maintains meeting minutes and documentation status; Zain updates development and defect status; Ahmad reviews the schedule, budget, risks, and client decisions in a weekly dashboard.

12. Communication & Meeting Planning:

- **Communication Methods:** WhatsApp for daily coordination, email for formal approvals, Google Meet for remote meetings, and shared Google Drive folders for controlled documents.

Meeting Type	Frequency	Participants	Agenda / Output
Daily Team Update	Working days	Atia, Ahmad, Zain	Current task, blocker, and next action

Meeting Type	Frequency	Participants	Agenda / Output
Team Meeting	Weekly	Project Team	Progress, risks, defects, schedule, and action list
Client Meeting	Every two weeks	Ahmad, Atia, Client	Requirement confirmation, prototype feedback, and approvals
Sprint Review	End of each sprint	Team and Client	Demonstration, acceptance feedback, and backlog update

13. Change Management Planning:

Process:

1. A stakeholder submits a change request with the reason and expected benefit.
2. Ahmad records the request and analyses its effect on scope, schedule, cost, quality, and risk.
3. Atia and Zain provide requirement and technical impact information.
4. Minor changes are approved by Ahmad; major changes require written client approval.
5. Approved changes are added to the change log, project baseline, backlog, and related documents.
6. The team implements and retests the approved change before closure.
 - **Minor Change:** A clarification or interface adjustment that does not increase the budget, alter the delivery date, or add a new major feature.
 - **Major Change:** A change that affects project scope, cost, architecture, database design, acceptance criteria, or final schedule.

14. RACI Matrix:

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Task	R	A	C	I
Requirement Gathering	Atia	Ahmad	Client, Zain	Team
System / UI Design	Atia, Zain	Ahmad	Client	Team
Database Design	Zain	Ahmad	Atia	Client
Product & Inventory Development	Zain	Ahmad	Atia	Client
Sales, Invoice & Reports Development	Zain	Ahmad	Atia	Client
Testing and UAT	Atia, Zain	Ahmad	Client	Team

Task	R	A	C	I
Deployment and Training	Ahmad, Zain	Ahmad	Atia, Client	Users
Documentation	Atia	Ahmad	Team	Client

15. Rollout Planning:

a) Deployment Strategy:

- Deploy the application first on a test computer using a copied database.
- Run smoke tests and UAT with the store owner and selected sales staff.
- Install the approved release on the store computer and import validated opening data.
- Keep the previous manual register available for one working day as a controlled fallback.

b) Production Environment:

- Windows desktop computer with Java Runtime Environment and MySQL 8 Community Server.
- Local or LAN-based database connection through JDBC.
- Daily database backup to an external drive with weekly restore verification.

c) Training Plan:

- Two-hour training session for the store owner, admin, and sales staff.
- Printed quick-start guide and step-by-step user manual.
- One-week post-deployment support for critical defects and usage questions.

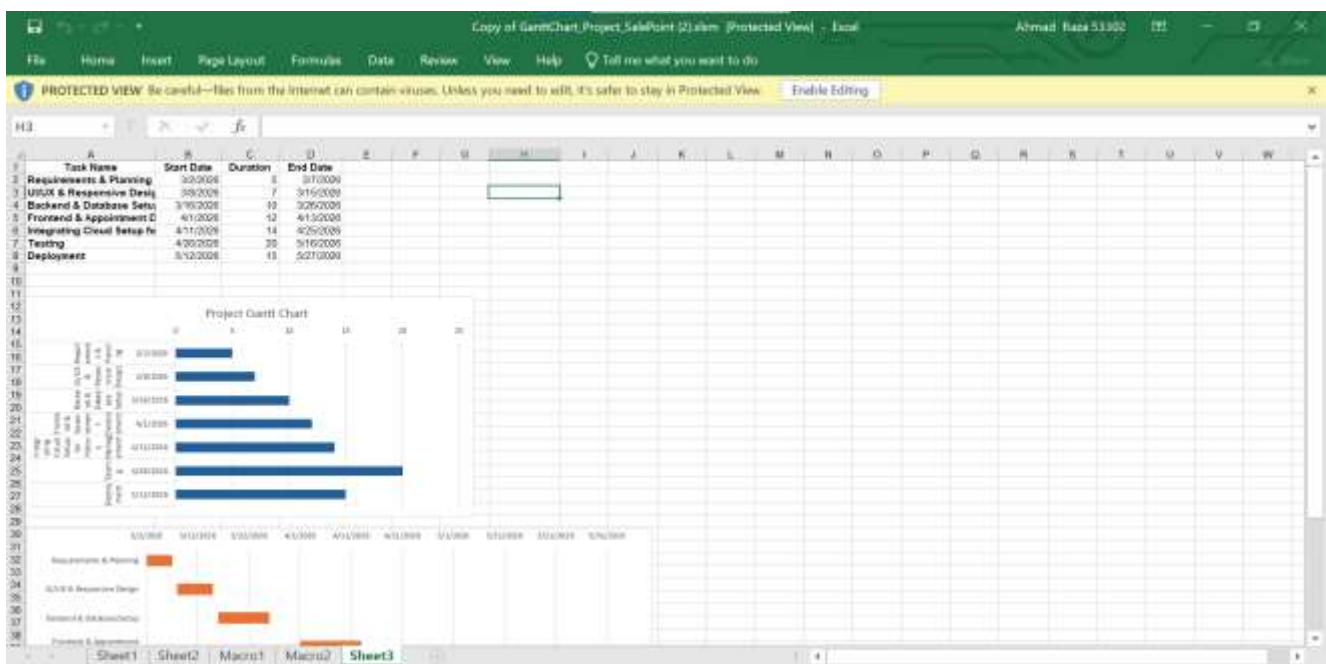
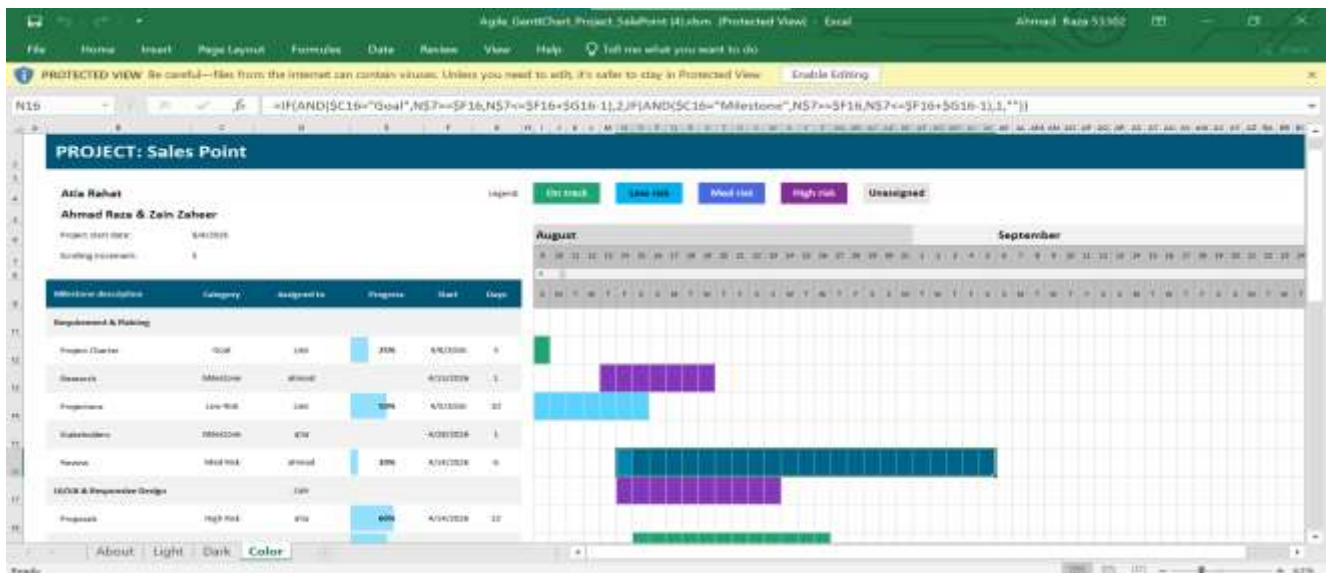
d) Acceptance Criteria:

- All agreed core modules operate according to approved requirements.
- At least 95% of planned test cases pass.
- Zero open critical or high-severity defects.
- Invoice generation completes within five seconds.
- Stock is updated within two seconds after a completed sale.
- Backup and restore procedures are successfully demonstrated.
- Client signs the UAT and final acceptance form.
- At least 90% positive feedback from at least five test users.

16. Gantt Chart:

Activity	W1	W2	W3	W4	W5	W6	W7	W8
Planning / Requirements	X							
Scope Approval	X							

Activity	W1	W2	W3	W4	W5	W6	W7	W8
System and UI Design		X						
Database Design		X						
Product / Inventory Dev.			X	X	X			
Sales / Invoice / Reports Dev.			X	X	X			
Integration						X		
Testing / UAT							X	
Deployment / Training								X
Documentation	X	X	X	X	X	X	X	X



17. MS Project:

The following project-specific visual represents the final SalesPoint schedule. It is a documentation mock-up and should be replaced by an exported MS Project screenshot if live-tool evidence is required.

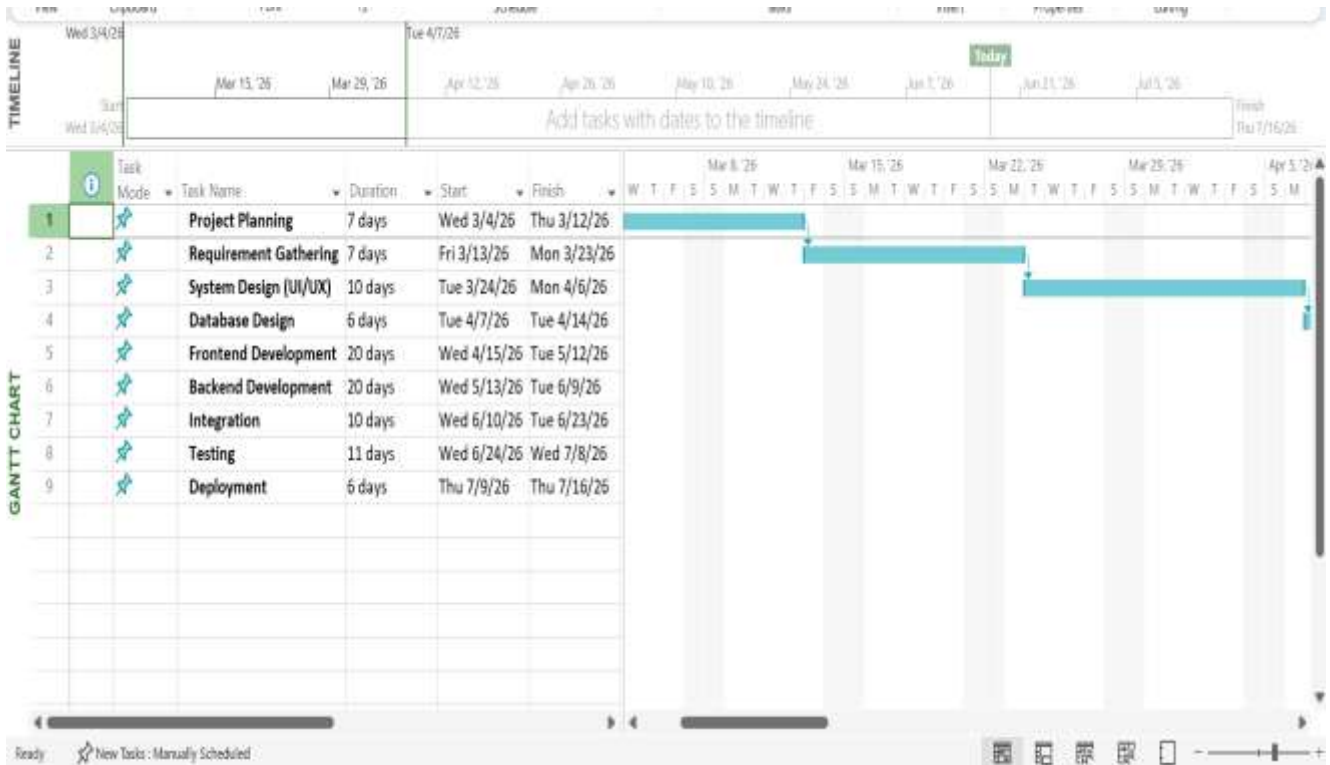


Figure 1: SalesPoint MS Project schedule view

19. MS Planner:

Dashboard / Board:

Tasks are grouped into Planning, Design, Development, and Testing & Rollout buckets. Each card shows an owner and planned week.

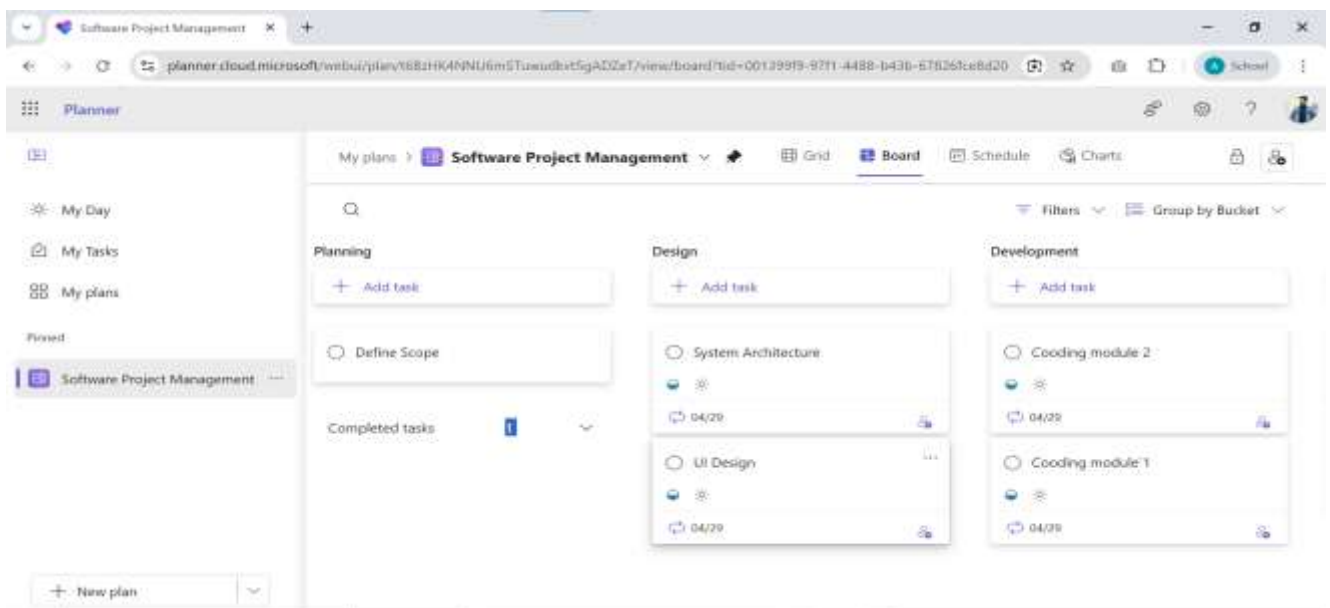


Figure 2: SalesPoint Planner board mock-up

Charts Dashboard:

The charts summarize total tasks, current status, and workstream distribution for weekly monitoring.

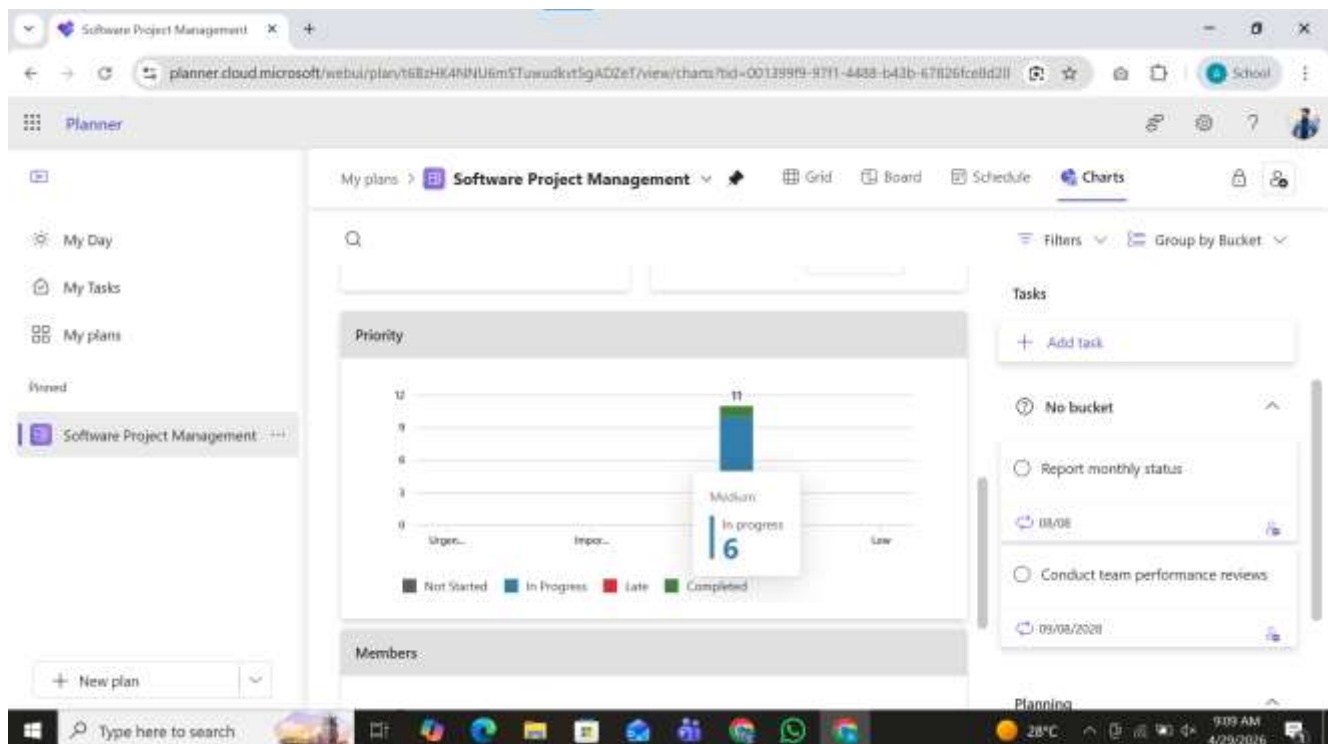
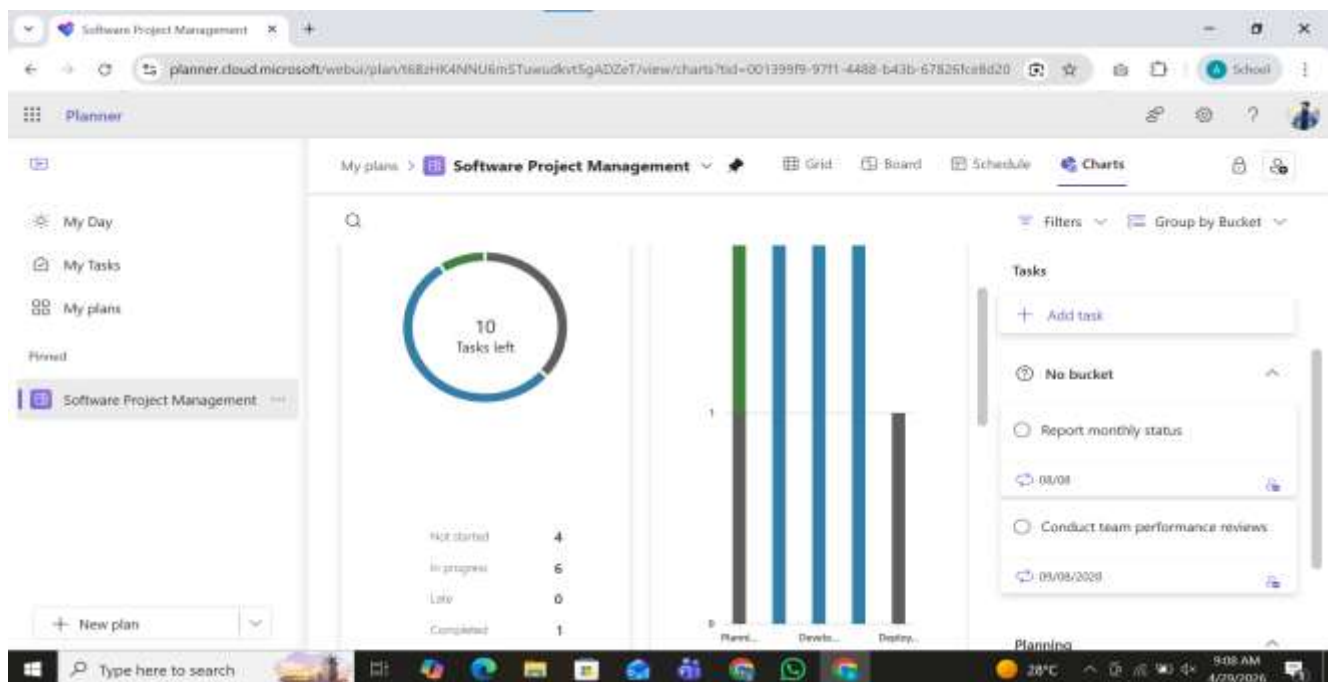


Figure 3: SalesPoint Planner charts mock-up

19. Taiga.io:

Team Members and Product Backlog:

The backlog uses proper user stories, priorities, story points, and sprint assignments. All three team members are shown as active.

Memberships - SPM Taiga lab | You've been Taigatized! - 5330 | Memberships - SPM Taiga lab |

tree.taiga.io/project/ahmad_07-spm-taiga-lab-project/admin/memberships

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PROJECT

ATTRIBUTES

MEMBERS

PERMISSIONS

INTEGRATIONS

PLUGINS

Manage members

+ NEW MEMBER

Member	Admin	Role	Status
 4 53812@students.riphah.edu.pk	☐	UX	Active 
 Ahmad Raza 53302@students.riphah.edu.pk	☐	Front	Active 
 57325@students.riphah.edu.pk	☐	Pending Design	 Resend

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Memberships - SPM Taiga lab | You've been Taigatized! - 5330 | Backlog - SPM Taiga lab project |

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project you have to set up the points and sprints through the Admin

Backlog

5 user stories

+ USER STORY

Filters subject or reference Tags

USER STORY	STATUS	POINTS
  #1 Requirement Gathering	New	?
  #2 system design	New	?
  #3 module development	New	?
  #4 testing	New	?
  #5 deployment	New	?

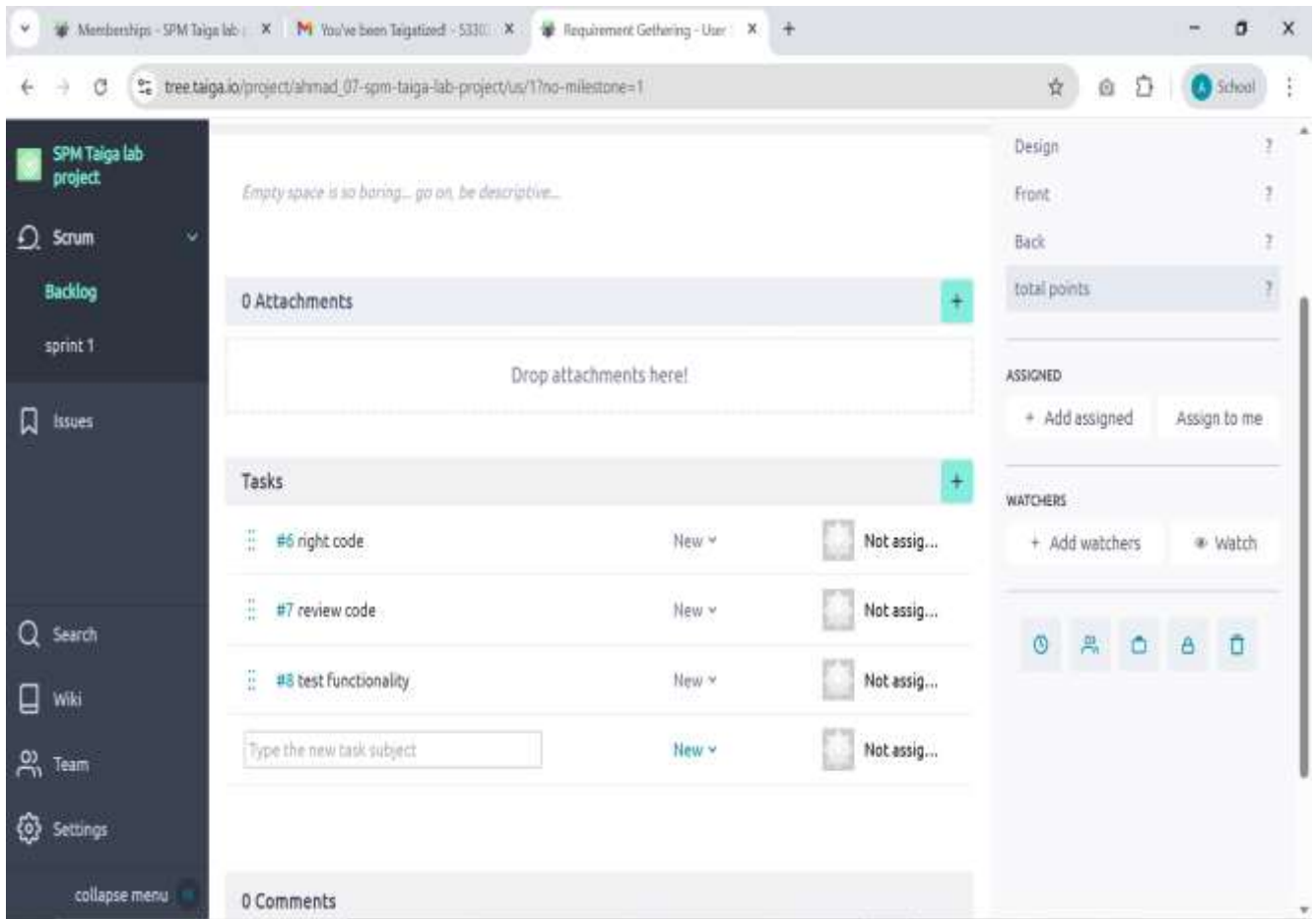
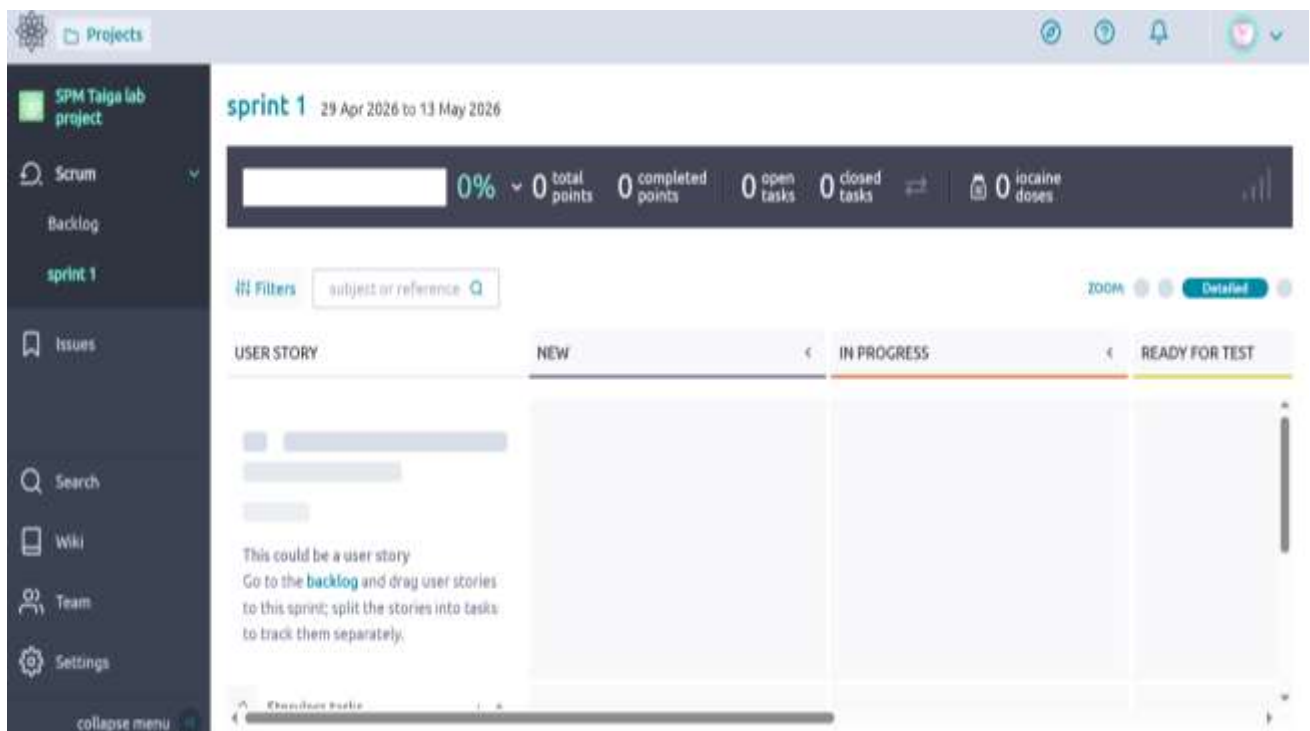


Figure 4: SalesPoint Taiga members and backlog mock-up

Sprint Taskboard:

Sprint tasks are assigned to named team members and moved through New, In Progress, Ready for Test, and Done states.



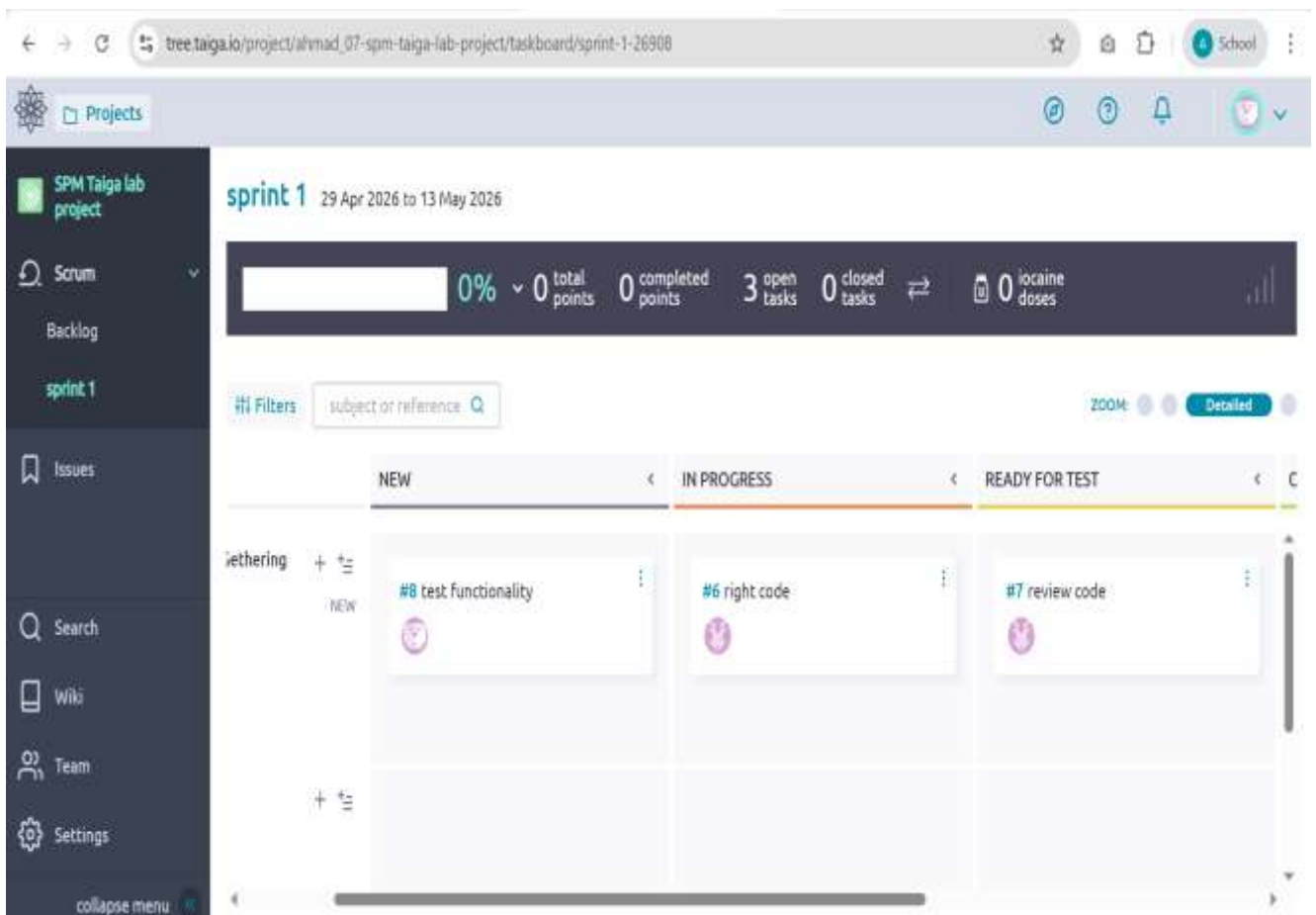
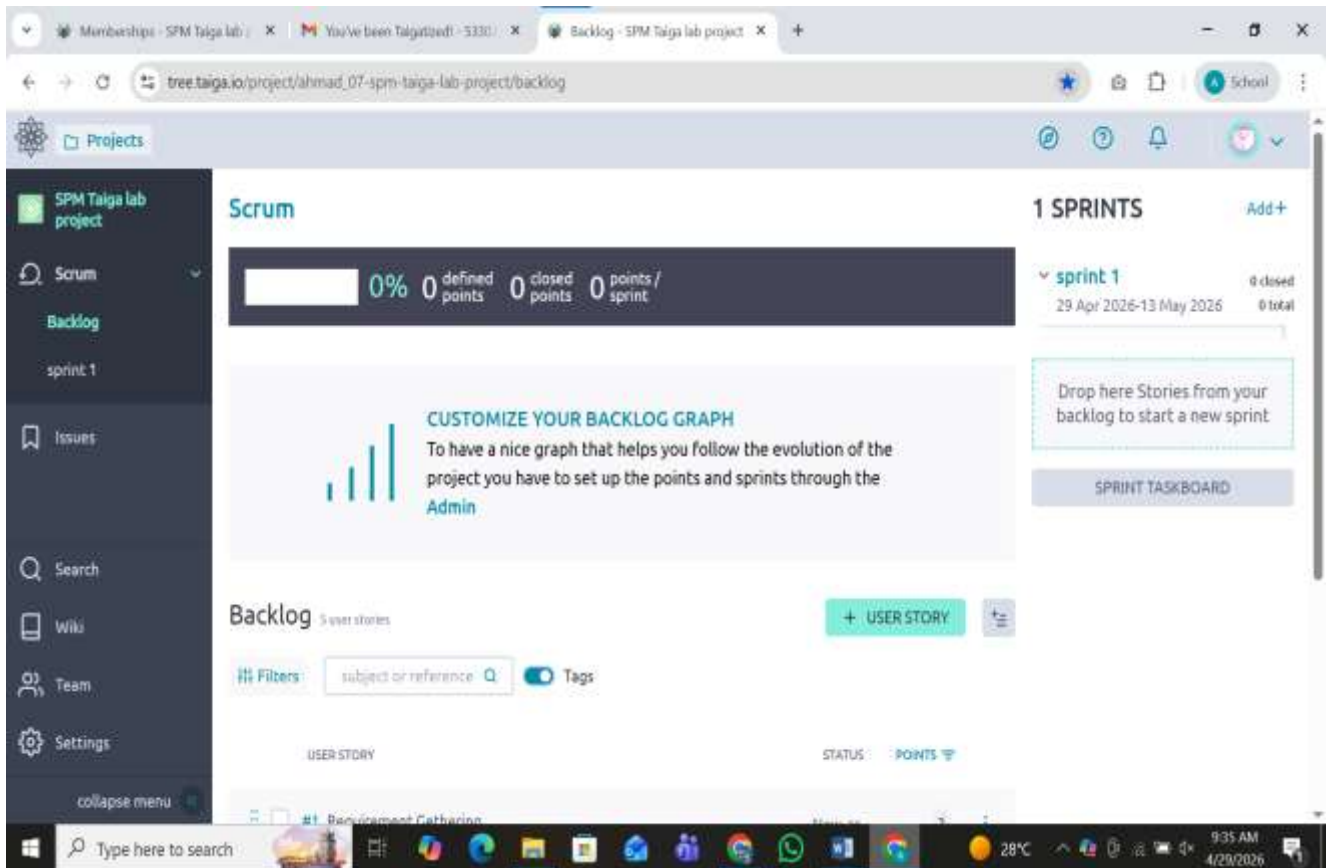


Figure 5: SalesPoint Taiga Sprint 1 taskboard mock-up

20. Tree Structure (Text-Based WBS):

- **SalesPoint (Sales and Inventory Management System)**

1. Inventory Management

- 1.1 Product Management
- 1.2 Stock Monitoring
- 1.3 Record Management

2. Sales Operations

- 2.1 Sales Transaction Processing
- 2.2 Invoice Generation
- 2.3 Billing Calculation

3. Reporting & Analytics

- 3.1 Daily and Monthly Sales Reports
- 3.2 Business Performance Analysis
- 3.3 Real-Time Business Insights

4. User Management

- 4.1 Admin Authentication
- 4.2 Sales Staff Access
- 4.3 Employee and Customer Records

5. Project Delivery

- 5.1 System Testing and UAT
- 5.2 Deployment and Backup
- 5.3 User Training and Documentation

21. Hand-Drawn WBS:

The corrected sketch places product and stock functions under Inventory and report functions under Reporting & Analytics.

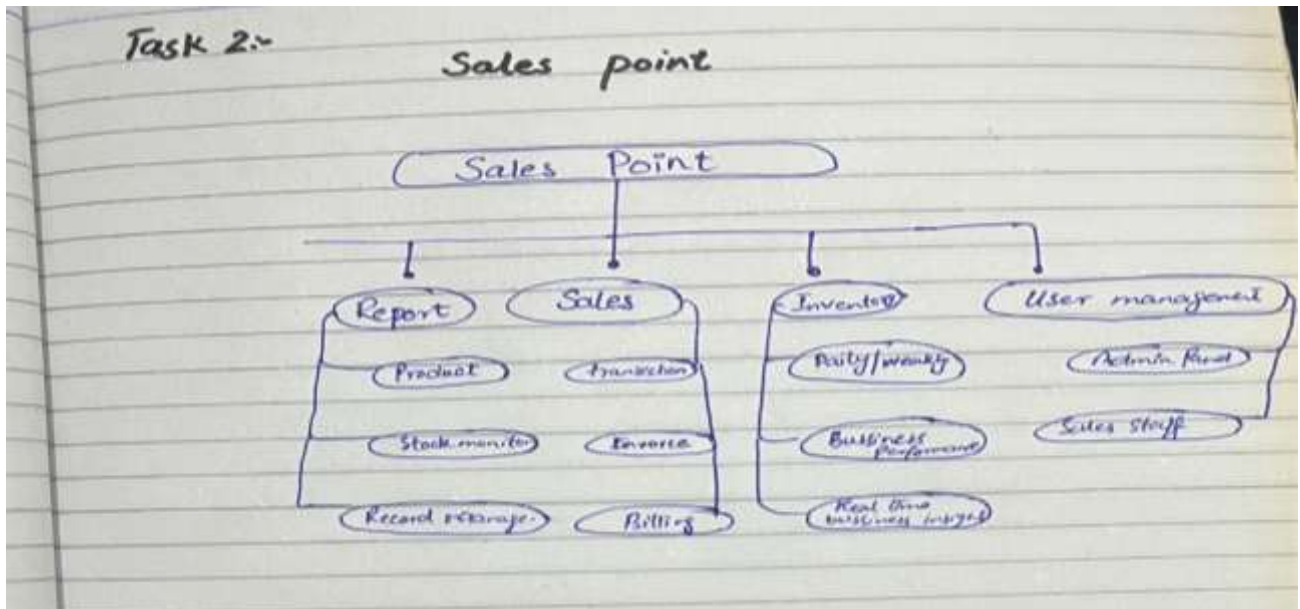


Figure 6: Corrected hand-drawn-style SalesPoint WBS

22. Draw.io WBS:



Figure 7: Corrected SalesPoint WBS diagram

23. Project Scheduling Using AON and CPM:

A software team is developing SalesPoint for a small general retail store. The following activities, durations, and dependencies are used to identify the critical path and minimum completion time.

Activity	Description	Duration (Days)	Predecessor
A	Requirements Gathering	5	—
B	Feasibility and Scope Approval	3	A
C	System and UI Design	5	B
D	Database Design	4	B
E	Product and Inventory Module Development	12	C

Activity	Description	Duration (Days)	Predecessor
F	Sales, Invoice and Reports Module Development	10	D
G	System Integration	5	E, F
H	System Testing and UAT	6	G
I	Deployment and User Training	4	H

SalesPoint AON Network Diagram (40 Working Days)

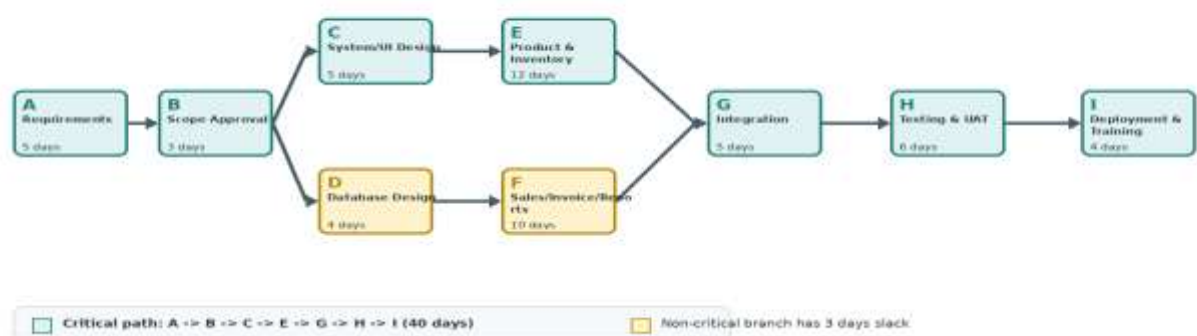


Figure 8: SalesPoint AON network diagram

Step 1: Forward Pass

Formula: $EFT = EST + \text{Duration}$. When an activity has multiple predecessors, its EST is the highest predecessor EFT.

Activity	EST	Duration	EFT
A	0	5	5
B	5	3	8
C	8	5	13
D	8	4	12
E	13	12	25
F	12	10	22
G	25	5	30
H	30	6	36
I	36	4	40

Step 2: Backward Pass

Formula: $LST = LFT - \text{Duration}$. When an activity has multiple successors, its LFT is the lowest successor LST.

Activity	LFT	Duration	LST
I	40	4	36
H	36	6	30
G	30	5	25
E	25	12	13
F	25	10	15
C	13	5	8
D	15	4	11
B	8	3	5
A	5	5	0

Step 3: Slack and Critical Path

Formula: $\text{Slack} = LST - EST$. Activities with zero slack belong to the critical path.

Activity	Duration	EST	EFT	LST	LFT	Slack
A	5	0	5	0	5	0
B	3	5	8	5	8	0
C	5	8	13	8	13	0
D	4	8	12	11	15	3
E	12	13	25	13	25	0
F	10	12	22	15	25	3
G	5	25	30	25	30	0
H	6	30	36	30	36	0
I	4	36	40	36	40	0

- **Critical Path:** A -> B -> C -> E -> G -> H -> I
- **Total Project Duration:** 40 working days (8 weeks)
- **Non-Critical Slack:** Activities D and F each have three working days of slack.

- **Result:** The CPM analysis confirms the same 40-working-day schedule used in the phase plan, Gantt chart, MS Project mock-up, and COCOMO capacity calculation.

24. Jira - Agile Project Management:

Sprint and task Board:

The board contains only SalesPoint issues and uses project key SP. Work is grouped into To Do, In Progress, In Review, and Done.

The figure displays two screenshots of the Jira interface for the SalesPoint project.

Top Screenshot: Sprint Board

The top screenshot shows the 'SalesPoint' project space with the 'Backlog' tab selected. A sprint titled 'SalesPoint Sprint 1' (15 Jun - 29 Jun) is active, containing 4 work items. The work items are listed as follows:

Issue Key	Summary	Status	Assignee	Reporter
SAL-11	T1: Create Database Schema and UI Wireframes	TO DO	ZZ	AR
SAL-13	T3: Finalize Stakeholder Needs and System Specs	TO DO	ZZ	AR
SAL-8	US-2: Manage and Update Products in Inventory	TO DO	Pyl	AR
SAL-7	US-1: Secure Admin Login Access	TO DO	AR	AR

Bottom Screenshot: All work Board

The bottom screenshot shows the 'All work' board for the SalesPoint project. It displays a list of work items with the following columns: Work, Assignee, Reporter, and a status icon. The work items are listed as follows:

Work	Assignee	Reporter	Status
SAL-16 Black-Box Testing	57325	AR AHMAD RAZA	M
SAL-15 T4: Perform System Integration and Black-Box Te...	57325	AR AHMAD RAZA	M
SAL-14 T4: Perform System Integration and Black-Box Te...	57325	AR AHMAD RAZA	M
SAL-13 T3: Finalize Stakeholder Needs and System Specs	57325	AR AHMAD RAZA	M
SAL-12 T2: Code the Core Sales and Inventory Modules	ZZ Zain Zaheer	AR AHMAD RAZA	M
SAL-11 T1: Create Database Schema and UI Wireframes	ZZ Zain Zaheer	AR AHMAD RAZA	M

Figure 9: SalesPoint Jira Sprint and task board mock-up

Epic and Story Board:

The backlog shows named assignees, epics, story points, and sprint placement. The report shows committed and completed points together with current test results.

The top screenshot shows the 'All work' board with 6 work items. The bottom screenshot shows the 'All work' board with 4 work items.

Work	Assignee	Reporter
☐ SAL-6 User Management	57325	AHMAD RAZA
☐ SAL-5 User Management	Zain Zaheer	AHMAD RAZA
☐ SAL-4 Reporting & Analytics	AHMAD RAZA	AHMAD RAZA
☐ SAL-3 Sales Operations	57325	AHMAD RAZA
☐ SAL-2 Sales Operations	AHMAD RAZA	AHMAD RAZA
☐ SAL-1 Inventory Management	57325	AHMAD RAZA

Work	Assignee	Reporter
☐ SAL-10 US-4: Automated Invoice Generation for Customers	AHMAD RAZA	AHMAD RAZA
☐ SAL-9 US-3: Process Sales Transactions and Automated ...	Zain Zaheer	AHMAD RAZA
☐ SAL-8 US-2: Manage and Update Products in Inventory	57325	AHMAD RAZA
☐ SAL-7 US-1: Secure Admin Login Access	57325	AHMAD RAZA

Figure 10: SalesPoint Epic and story board

Scrum Board:

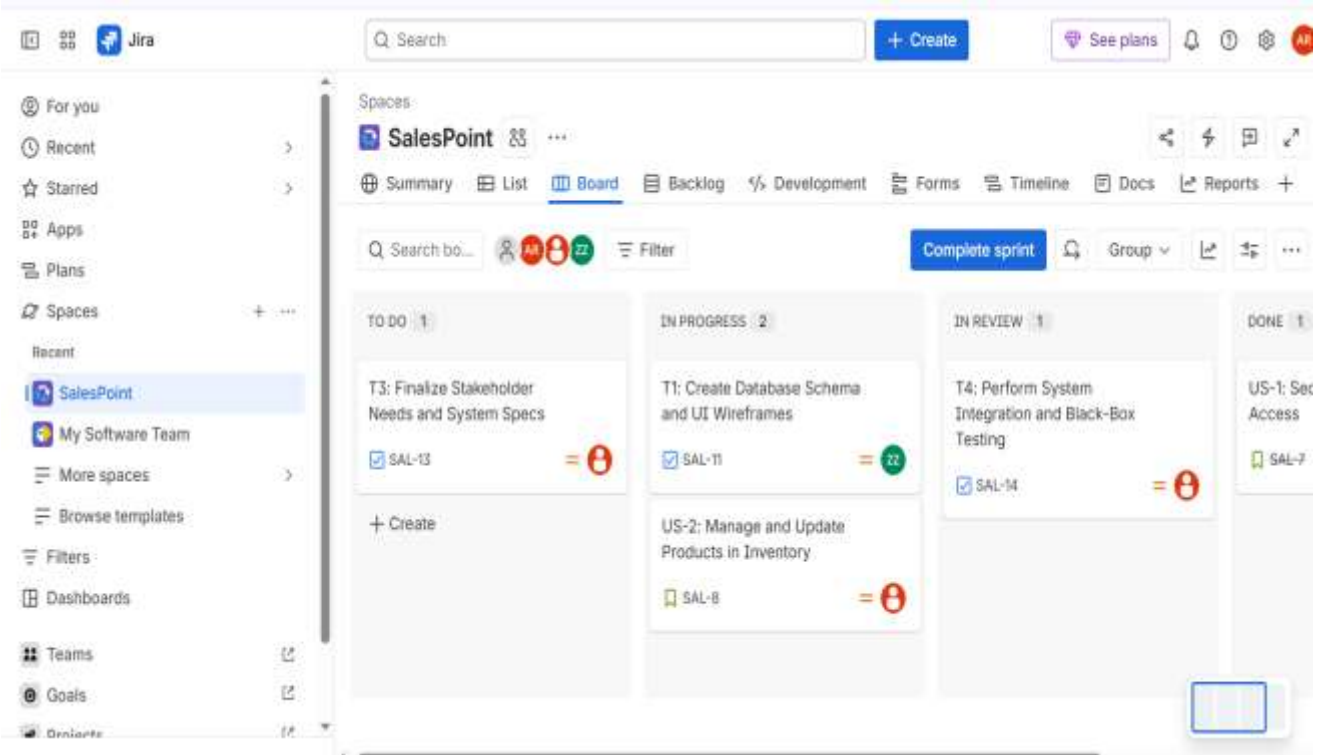


Figure 11: SalesPoint Scrum board

Bugs:

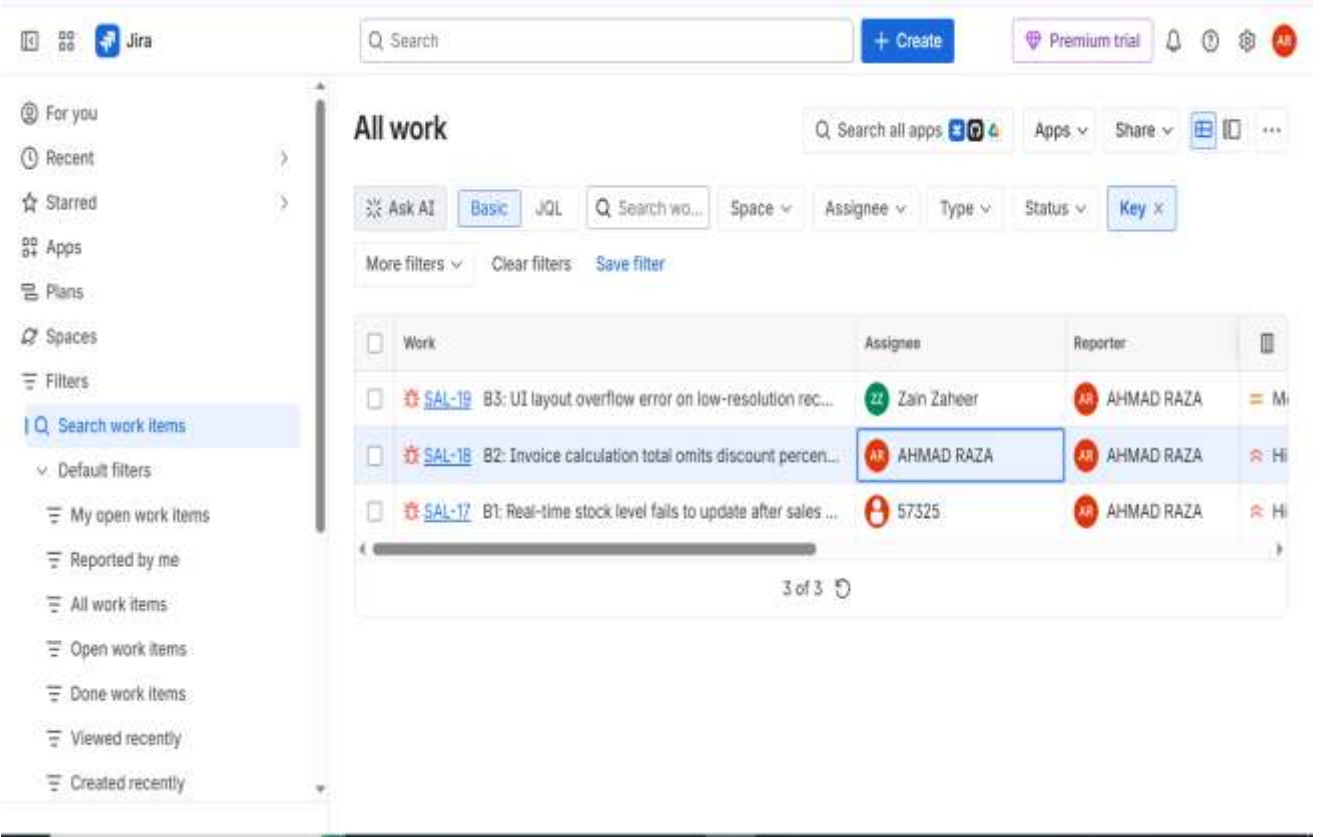


Figure 12: SalesPoint Bugs board

25. LOC Estimation for SalesPoint:

Step 1: Estimated Modules

Module	Estimated LOC
User Authentication & Login	200
Product Management	400
Inventory Management	350
Sales Transaction Processing	550
Invoice Generation	250
Reports & Analytics	300
Customer Management	150
Employee Management	200
Total LOC	2,400

Step 2: Calculate Total LOC

$$\text{Total LOC} = 200 + 400 + 350 + 550 + 250 + 300 + 150 + 200 = 2,400 \text{ LOC}$$

Step 3: Convert to KLOC

$$\text{KLOC} = \text{LOC} / 1,000 = 2,400 / 1,000 = 2.4 \text{ KLOC}$$

- **Estimation Basis:** The estimate represents a controlled academic prototype using Java Swing libraries, reusable JDBC utilities, and a limited single-store scope.

Metric	Value
Total LOC	2,400
KLOC	2.4

26. Function Point Analysis (FPA):

Step 1: Identify Components

Component	Count	Low-Complexity Weight	Total
External Inputs (EI)	7	3	21
External Outputs (EO)	5	4	20
External Queries (EQ)	5	3	15

Component	Count	Low-Complexity Weight	Total
Internal Logical Files (ILF)	5	7	35
External Interface Files (EIF)	0	5	0

Component Details

- **Inputs (7):** User login, add product, update product, delete product, record sale, add customer, and add employee.
- **Outputs (5):** Invoice, daily sales report, monthly sales report, stock report, and employee/customer report.
- **Queries (5):** Search product, check stock availability, view sales history, search customer, and search employee.
- **Internal Files (5):** User, Product/Inventory, Sales, Customer, and Employee logical data groups.
- **External Interface Files (0):** The MySQL database is maintained by SalesPoint itself. The receipt printer is an output device, not an external logical file.

Step 2: Calculate UFP

$$\text{UFP} = 21 + 20 + 15 + 35 + 0 = 91$$

Step 3: Apply Value Adjustment Factor

For this controlled academic prototype, $\text{VAF} = 1.00$. $\text{Final FP} = 91 \times 1.00 = 91$.

Metric	Value
Unadjusted Function Points	91
Final Function Points	91

27. COCOMO Estimation:

SalesPoint is a small, familiar, and relatively stable desktop business application developed by a small team; therefore, the Basic COCOMO Organic mode is used for an academic effort estimate.

Constant	Value
a	2.4
b	1.05

Effort Formula

$$\text{Effort} = 2.4 \times (\text{KLOC})^{1.05}$$

$$\text{Effort} = 2.4 \times (2.4)^{1.05} = \text{approximately } 6.02 \text{ person-months}$$

Schedule Consistency Check

Planned team capacity = 3 team members x 2 months = 6 person-months. Therefore, the eight-week project schedule is consistent with the COCOMO effort estimate for the controlled prototype.

Metric	Estimated Value
LOC	2,400
KLOC	2.4
Function Points	91
Estimated Effort	6.02 person-months
Team Size	3 members
Planned Duration	Approximately 2 months / 8 weeks

Factors Affecting SalesPoint

Factor	Effect on Project
Team Experience	Familiarity with Java, Swing, SQL, and testing reduces learning time.
Product Complexity	Real-time stock deduction and accurate invoices require careful validation.
Reliability	Sales totals, stock quantities, and database transactions require thorough testing.
Tools & Reuse	Swing components, JDBC utilities, and reusable validation code reduce LOC and effort.
Schedule Constraint	The eight-week limit requires scope control and weekly monitoring.
Data Security	Authentication, database access control, backup, and restore procedures add effort.

Final Conclusion:

The updated plan uses one consistent scope, one Java Swing and MySQL technology stack, one three-person team, one eight-week schedule, and matching LOC, Function Point, and COCOMO estimates. The project-management visuals also contain SalesPoint-specific tasks instead of unrelated project data.